

Technical Summary Sheet

Phase-Adaptive Quantization: DSE of 4-Bit MMA Rescale Pipelines

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Repository: <https://github.com/ET823828/Hardware-for-quantization>

Problem in one sentence

NVFP4 / MXFP4 / INT4 each bundle a single fixed rescale pipeline that is applied to both prefill and decode, even though the two phases have opposite hardware bottlenecks; for a custom AI accelerator the rescale pipeline is itself a design-space variable that should be co-optimized with the workload and the inference phase.

What we built

A 10-configuration W4A4 search space (**C0–C9**) decomposed across three orthogonal axes—rescale levels, fine block size b , and (scale $\times$ accumulator) precision—containing MXFP4 (**=C1**) and NVFP4 (**=C7**) as embedded points. The full pipeline is formalized as seven canonical Einsum operators (**BlockScaleA**, **BlockQuantA**, **MatMulBlock**, **RescaleBlockA/W**, **RescaleTensorA/W**); every configuration is a subset of these operators with per-level scale and accumulator precision, evaluated by AccelForge wrapping Timeloop+Accelergy on a parameterized 8×8 PE-array accelerator (64KB GLB, 128B RF/PE, 45nm). Sweep = $10\times 3\times 2 = 60$ hardware runs + 60 matched accuracy-emulation runs across LLM Llama-2 FFN, VLM Qwen2-VL ViT, VLA OpenVLA action head $\times$ {prefill ($M=128$), decode ($M=1$)}, with a $\cos \geq 0.98$ accuracy floor.

Three key insights

(1) NVFP4 is dominated everywhere. The C7 NVFP4-like reference is *never* the lowest-energy configuration that clears the cosine floor in any of the 6 (workload, phase) cells. Every selected best replaces FP32 rescale with FP16 or E8M0 with < 0.005 cosine drop, saving 48–60% energy. Adopting NVFP4 as-is into a custom W4A4 datapath leaves $\sim 2\times$ energy on the table.

(2) A two-step efficient search procedure for the W4A4 design space. *Step 1*—the cosine floor admits a block-granularity family ($b=32$ if its only member C1 passes, else $b=16$). *Step 2*—within the admitted family, pick the configuration with the lowest closed-form per-output cost at the cell’s (M, N, K) . On all six (workload, phase) cells the procedure returns the same configuration as exhaustive sweep at $\mathcal{O}(1)$ accuracy probes per cell; whether it generalises beyond our three workload classes is a hypothesis for future work.

(3) Phase-adaptive value is conditional, with VLA the clearest beneficiary. Versus fixed-NVFP4, phase-adaptive saves 50–60% across all three workloads. But versus the best single fixed configuration the gain is $\approx 0\%$ for VLM, 0.6% for LLM, and 8.2% for VLA at $\alpha=0.5$: VLA is the only workload in our sweep that warrants paying silicon area for a second rescale mode; VLM should ship a single MXFP4-like mode. An analytic per-output cost decomposition shows the C7-vs-best gap is dominated by activation-side fine rescale terms that are independent of M , so the headline saving is monotone non-decreasing in M and our $M=128$ profiling point is conservative.

Headline numbers

Workload	Phase	Best	pJ/output	Cosine	vs. NVFP4 (C7)
LLM	Prefill	C1	5 262	0.984	−58.1%
LLM	Decode	C8	7 767	0.981	−48.0%
VLM	Prefill	C1	3 952	0.984	−60.0%
VLM	Decode	C1	5 557	0.987	−50.4%
VLA	Prefill	C9	6 263	0.990	−50.7%
VLA	Decode	C8	7 891	0.995	−47.6%

Partner contributions

Yichong Zhang: Idea formulation, Einsum formalization, hardware-side AccelForge sweep harness, Pareto and phase-adaptive analysis pipeline, paper writing.

Wenye Xiong: quantization emulator, tensor-snapshot preparation for LLM/VLM/VLA workloads, accuracy sweep automation, accuracy floor analysis, repo cleaning.

Joint: search-space definition, figures, presentation slides, posters.

Reproducibility

Please follow the instructions under the repo.